

11.1

Factors and Levels Identify the treatments or factors and levels in Exercises 8–11.

8. A grower wishes to compare three types of fertilizers as they affect crop yield.

factor = type of fertilizer

levels = the 3 types of fertilizers

9. A researcher wishes to investigate the effect of daily doses of vitamin C at doses of 200, 500, and 1000 mg on preventing the common cold.

factor = daily dose of vitamin C

levels = 200 mg, 500 mg, 1000 mg

Normality? Explain why you would or why you would not be willing to assume that the data described in Exercises 12–16 are normal.

14. The proportion of defective items among 100 items selected from daily output is recorded for 25 days.

not normal. The data are proportions based on counts out of 100 items, so the values are bounded btw 0 and 1. Usually binomial rather than normally distributed.

111 2

DATA
SET

DS1103

8. Reducing Hostility A psychologist wants to compare three methods for reducing hostility levels in university students. Eleven students were judged to have "great hostility," based on a certain psychological test (HLT). These students were then randomly divided into three groups—five were treated by method A, three were treated by method B, and the other three students were treated by method C. All treatments continued throughout a semester, at the end of which the HLT test was given again. The results are shown in the table.

Method	Scores on the HLT Test				
A	73	83	76	68	80
B	54	74	71		
C	79	95	87		

- Perform an analysis of variance for this experiment.
- Do the data provide sufficient evidence to indicate a difference in mean student scores after treatment for the three methods?

a1 ANOVA

df 55 MS F

treatment 2 6551.52 327.76 4.11

error 8 638.3 79.79

total 10 1293.82

b1

$$F_{0.05, 2, 8} = 4.46$$

$$F = 4.11 < 4.46$$

Fail reject H_0 , no sufficient evidence

10. Assembling Electronic Equipment An experiment was conducted to compare the effectiveness of three training programs, A, B, and C, for assemblers of a piece of electronic equipment. Five employees were randomly assigned to each of three programs. After completion of the program, each person assembled four pieces of the equipment, and their average assembly time was recorded. Several employees resigned during the course of the program; the remainder were evaluated, producing the data shown in the accompanying table. Use the *Excel* printout to answer the questions in parts a–d.

Training Program	Average Assembly Time (min)				
A	59	64	57	62	
B	52	58	54		
C	58	65	71	63	64

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
A	4	242	60.5	9.667
B	3	164	54.667	9.333
C	5	321	64.2	21.7

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	170.45	2	85.225	5.704	0.0251	4.256
Within Groups	134.467	9	14.941			
Total	304.917	11				

- Do the data indicate a significant difference in mean assembly times for people trained by the three programs? Give the p -value for the test and interpret its value.
- Find a 99% confidence interval for the difference in mean assembly times between persons trained by programs A and B.
- Find a 99% confidence interval for the mean assembly times for persons trained by program A.
- Do you think the data will satisfy (approximately) the assumption that they have been selected from normal populations? Why?

a1	df	SS	MS	F
treatment	2	170.45	85.225	5.704
error	9	134.467	14.941	
total	11	304.917		

$p\text{-value} = 0.0251 < 0.105$, 有显著差异

$$b1 \quad \bar{X}_A = 60.5, \quad \bar{X}_B = 54.667$$

$$60.5 - 54.667 = 5.833$$

$$5.833 \pm 9.594$$

$$(-3.761, 15.427) \neq$$

C1 (56, 957, 6410437)

d. No, sample size too small.

DATA SET

DS1108

14. Tuna Fish The estimated average prices for a variety of brands of tuna fish, based on prices paid nationally, are shown here.¹

Light Tuna in Water	White Tuna in Water	White Tuna in Oil	Light Tuna in Oil
.99	.53	1.49	1.27
1.92	1.41	1.29	1.22
1.23	1.12	1.27	1.19
.85	.63	1.35	1.22
.65	.67	1.29	
.69	.60	1.00	
.60	.66	1.27	
		1.28	

- Construct the ANOVA table for a completely randomized design.
- Is there evidence of a significant difference in average price for these packages at the $\alpha = .05$ level of significance? At the $\alpha = .01$ level of significance?
- Find a 95% confidence interval estimate of the difference in price between light tuna in water and light tuna in oil.
- Find a 95% confidence interval estimate of the difference in price between white tuna in water and white tuna in oil.
- What other confidence intervals might be of interest to the researcher who conducted this experiment?

a.

	df	SS	MS	F
treatment	3	4.796	1.599	8.43
error	22	4.172	0.19	
total	25	8.968		

b. $F_{0.05, 3, 22} = 3.105$

$F_{0.01, 3, 22} = 4.82$

$8.43 > 3.105$ $\text{rej } H_0$

$8.43 > 4.82$

c) $(-0.184, 0.104)$

d) $(-0.136, 0.146)$

in water

e) light tuna vs white tuna
light tuna vs, white tuna
in oil

11.3

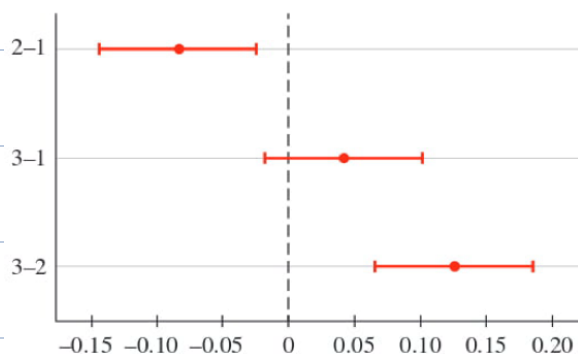
14. Calcium, again Refer to Exercise 13 (Section 11.2), where the calcium content of a powdered mineral substance was analyzed using three different methods. The comparisons option in MINITAB provided the printout that follows.

Tukey Simultaneous Test for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
2-1	-0.0840	0.0224	(-0.1438, -0.0242)	-3.75	0.007
3-1	0.0420	0.0224	(-0.0178, 0.1018)	1.87	0.189
3-2	0.1260	0.0224	(0.0662, 0.1858)	5.62	0.000

Individual confidence level = 97.94%

Tukey Simultaneous 95% CIs
Differences of Means for Calcium



If an interval does not contain zero, the corresponding means are significantly different.

Use Tukey's method to compare the three population means. How would you summarize your results?

1) 2-1

$(-0.1438, -0.0242)$

後也含0, 有顯著差異

② < ①

2) 3-1 $(-0.0178, 0.1018)$

No significantly difference

$$31 \quad 3-2 \quad (010662, 011858)$$

$$\textcircled{3} > \textcircled{2}$$

16. The Cost of Lumber, continued The analysis of variance F -test in Exercise 15 (Section 11.2) indicated that there was a difference in the average cost of lumber for the four states. The following information from that exercise is given in the table:

Sample Means $\bar{x}_1 = 262.2$	MSE	41.25
$\bar{x}_2 = 234.8$	Error df :	16
$\bar{x}_3 = 251.6$	n_j :	5
$\bar{x}_4 = 268.6$	k :	4

Use Tukey's method for paired comparisons to determine which means differ significantly from the others at the $\alpha = .01$ level.

$$q_{0.01, 4, 16} = 5.119$$

$$HSD = 5.119 \sqrt{\frac{41.25}{5}} = 14.91$$

x_2 和其他者都有顯著差異

x_3, x_4 有

x_1 和 x_3, x_4 沒 #

17. GRE Scores The quantitative reasoning scores on the Graduate Record Examination (GRE)³

were recorded for students admitted to three different graduate programs at a local university.

Graduate Program					
Life Sciences		Physical Sciences		Social Sciences	
630	660	660	760	440	540
640	660	640	670	330	450
470	480	720	700	670	570
600	650	690	710	570	530
580	710	530	450	590	630

- Do these data provide sufficient evidence to indicate a difference in the mean GRE scores for applicants admitted to the three programs?
- Find a 95% confidence interval for the difference in mean GRE scores for Life Sciences and Physical Sciences.
- If you find a significant difference in the average GRE scores for the three programs, use Tukey's method for paired comparisons to determine which means differ significantly from the others. Use $\alpha = .05$.

	df	SS	MS	F
treatment	2	74806.67	37403.33	4.47
error	27	225930	8367.78	
total	29	300736.67		

$$p = 0.021 < 0.05, \text{ rej. } H_0 \#$$

$$b) \bar{x}_1 = 608, \bar{x}_2 = 653$$

$$95\% \text{ CI} = (-1281.94, 381.94)$$

$$CI \quad HSD = 10.143$$

$$\bar{A} = 608$$

$$\bar{B} = 653$$

$$\bar{C} = 532$$

Physical sci | social

sci 有顯著差異 #